

PROJECT REPORT

Smart Personal Finance & Goal Management System

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MID SEMESTER EVALUATION

UCS503 – SOFTWARE ENGINEERING

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Chapter 1

Abstract

The Smart Personal Finance & Goal Management System is a web-based application designed to help users manage income, expenses, budgets and financial goals in one place. The system combines transaction management with rule-based financial intelligence, anomaly detection, forecasting and scenario simulation.

The proposed system follows a modular architecture with a React/Tailwind frontend, a REST API backend using Java and a MySQL database. The analytics layer uses transaction history to identify spending patterns, recurring behaviour and unusual transactions. A rule engine converts financial conditions into understandable alerts and recommendations, while Monte Carlo simulation is used for goal and scenario projections.

The main objective is to provide users with a clear dashboard, structured financial records and explainable insights that can support better personal financial planning.

Chapter 2

Introduction and Problem Statement

Managing personal finances often involves multiple disconnected activities such as recording transactions, checking account balances, setting budgets, tracking savings goals and evaluating future financial plans. A user may therefore have the data but still lack a simple way to understand spending behaviour and the effect of different financial decisions.

The proposed Smart Personal Finance & Goal Management System addresses this problem by combining core personal-finance management with automated analysis. The system provides a single interface for transactions, categories, budgets and goals and extends these features with behaviour analysis, anomaly detection, a rule engine, forecasting and scenario comparison.

2.1 Problem Statement

Users need a simple and structured system to:

- record and categorize income and expenses;
- monitor balances and spending patterns;
- create and track budgets;
- create financial goals and monitor their progress;
- identify unusual spending behaviour;
- understand the reasons behind financial alerts;
- compare possible financial scenarios; and
- receive understandable recommendations based on their financial data.

2.2 Objectives

The main objectives are:

1. Develop secure user registration and login using JWT authentication.
2. Provide transaction creation, viewing, updating, deletion and categorization.
3. Provide budget creation and monitoring.
4. Provide financial goal creation and progress tracking.
5. Generate monthly, category-wise and trend-based spending summaries.
6. Detect unusual or anomalous spending behaviour.
7. Implement a configurable rule engine for financial alerts.
8. Implement forecasting and Monte Carlo based scenario simulation.
9. Compare financial scenarios and generate explainable recommendations.
10. Provide a clean dashboard for balances, spending, goals and insights.
11. Test the application using unit, integration and property-based testing.

2.3 Scope of the System

2.3.1 Core MVP Scope

The minimum viable product consists of:

- Register / Login
- Dashboard
- Add and manage transactions
- Transaction categorization
- Budget management
- Goal creation and tracking

2.3.2 Advanced Scope

The advanced feature flow consists of:

- Transaction history analysis
- Behaviour analysis
- Anomaly detection
- Rule engine
- Monte Carlo simulation
- Scenario comparison
- Smart recommendations

Chapter 3

System Workflow

The overall user workflow is:

**Register/Login → Dashboard → Add Transaction → Categorize Transaction
→ Create Goal → Track Goal**

The intelligent-analysis workflow is:

**Transaction History → Behaviour Analysis → Anomaly Detection → Rule
Engine → Monte Carlo Simulation → Scenario Comparison →
Recommendation**

Chapter 4

Technology Stack

Layer	Technology / Tools
Frontend	React, Tailwind CSS
Backend	REST API, Java
Database	MySQL; synthetic financial data for the ML/analytics part
Authentication	JWT
Data Analysis	Pandas, NumPy
Testing	Pytest, Hypothesis, integration and property-based tests
DevOps	Docker, GitHub Actions
Documentation	API documentation, README and project report

Table 4.1: Technology stack

Chapter 5

Functional Modules

5.1 Authentication Module

The authentication module supports registration and login. JWT-based authentication is used so that authenticated users can access protected financial data.

5.2 Transaction Module

Users can add, view, update and delete transactions. Transactions can be categorized so that spending can later be analyzed by category and time period.

5.3 Budget Module

Users can define budgets for relevant spending categories and monitor whether their actual spending remains within the selected limits.

5.4 Goal Module

Users can create financial goals and track progress towards them. Goal information is also used by the analysis and scenario modules.

5.5 Analytics Module

The analytics layer calculates financial metrics such as monthly spending, category-wise spending, average spending, savings and spending trends. It also identifies recurring patterns.

5.6 Anomaly Detection

The system checks transaction and spending behaviour for unusual changes. For example, an increase in spending beyond a defined threshold can be treated as a possible anomaly.

5.7 Rule Engine

The rule engine converts financial conditions into explainable actions. Example rules include:

- If Food spending $>$ Food budget, generate a warning.
- If spending increases by more than 20%, detect an anomaly.
- If goal probability $<$ 70%, suggest increasing savings.

Users can view, edit, enable or disable rules.

5.8 Scenario Simulation

The system can compare possible financial scenarios. Monte Carlo simulation can produce probability ranges such as P10, P50 and P90 projections for future outcomes.

5.9 Recommendation Module

Recommendations are generated from financial metrics, rules, anomalies and scenario results. The system is intended to make recommendations explainable by showing the relevant rule or financial condition.

Chapter 6

Database and API Design

6.1 Database Design

The proposed database contains the following major tables:

Table	Purpose
users	User account and authentication information
accounts	Financial account information
transactions	Income and expense records
categories	Transaction categories
budgets	User budget information
goals	Financial goals and progress
rules	Configurable financial rules
anomalies	Detected unusual spending events
scenarios	Saved financial scenarios
households	Household-level financial grouping
household_members	Users belonging to a household
approvals	Approval records for relevant household actions
audit_logs	Records of important system actions

6.2 API Design

Representative API endpoints include:

Endpoint	Purpose
POST /auth/register	Register a new user
POST /auth/login	Authenticate a user
GET /transactions	Retrieve transactions
POST /transactions	Add a transaction
PUT /transactions	Update a transaction
DELETE /transactions	Delete a transaction
GET /goals	Retrieve goals
POST /goals	Create a goal
PUT /goals	Update a goal
GET /budgets	Retrieve budgets
POST /budgets	Create a budget

Table 6.2: Representative API endpoints

Chapter 7

Frontend and Security

7.1 Frontend Screens

The proposed frontend contains:

- Login / Register
- Dashboard
- Transactions
- Budget
- Goals
- Scenario Planning
- Smart Insights
- Rules
- Profile

The dashboard presents balance, income, expenses, savings, spending charts, goal progress, smart recommendations and anomaly alerts.

7.2 Security and Data Handling

The system includes:

- JWT authentication;
- user-level access control;

- input validation;
- protected backend endpoints;
- secure handling of financial information; and
- audit logging for relevant system actions.

Chapter 8

UML Diagrams

8.1 Collaboration Diagram

Collaboration Diagram – Smart Personal Finance & Goal Management

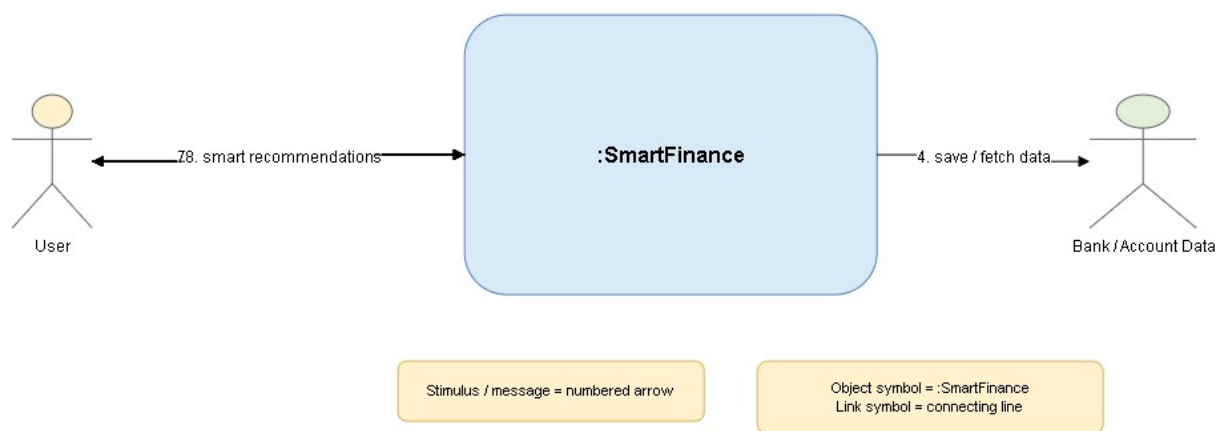


Figure 8.1: Collaboration Diagram

8.2 Sequence Diagram

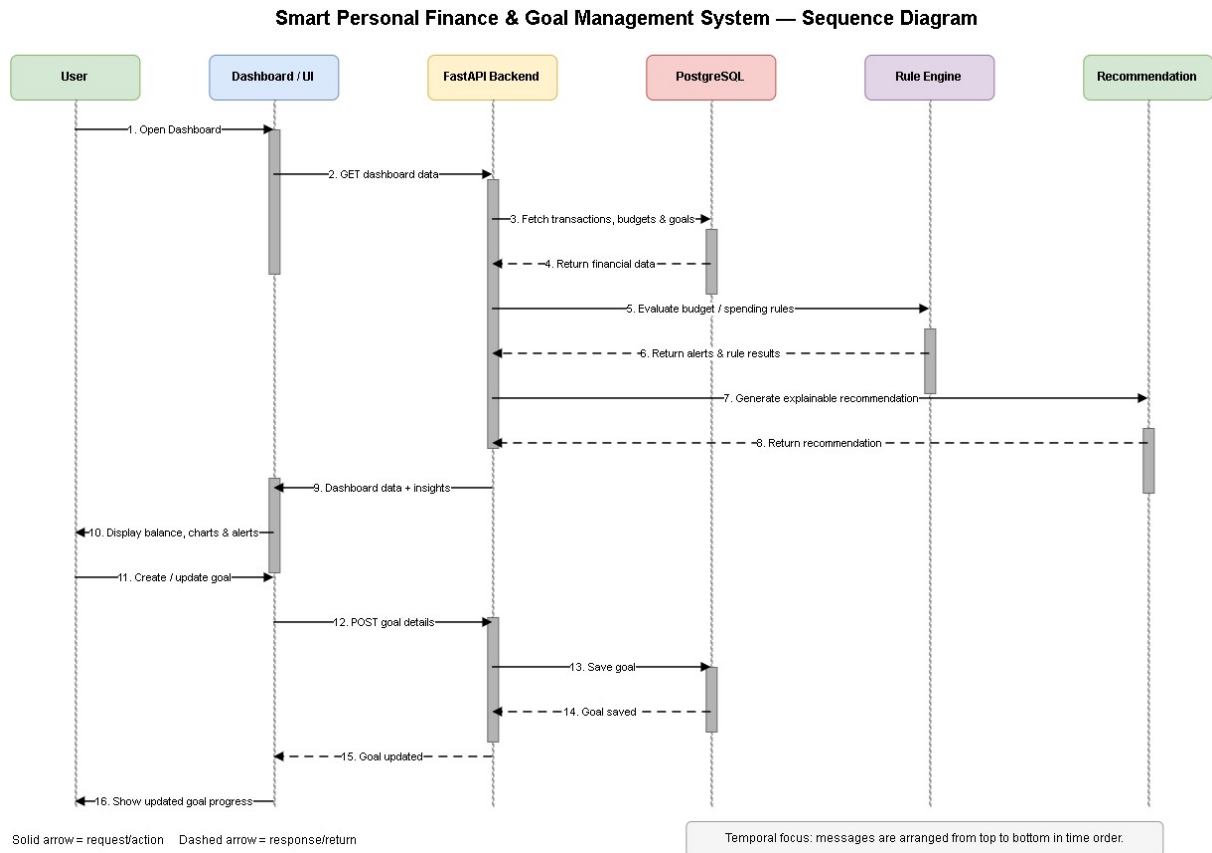


Figure 8.2: Sequence Diagram – Smart Personal Finance & Goal Management System

8.3 Activity Diagram

Activity Diagram – Smart Personal Finance & Goal Management

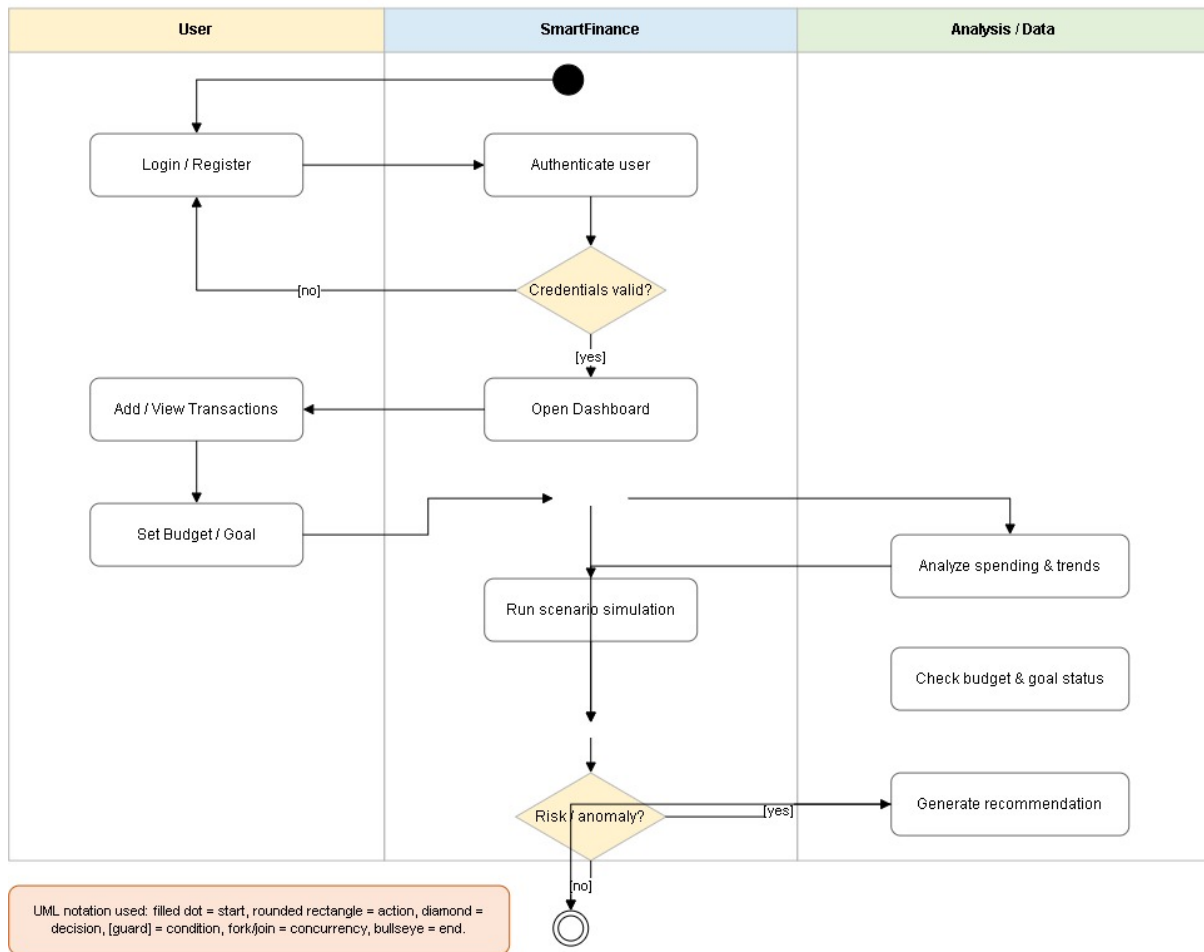


Figure 8.3: Activity Diagram

8.4 Swimlane / Partition Activity Diagram

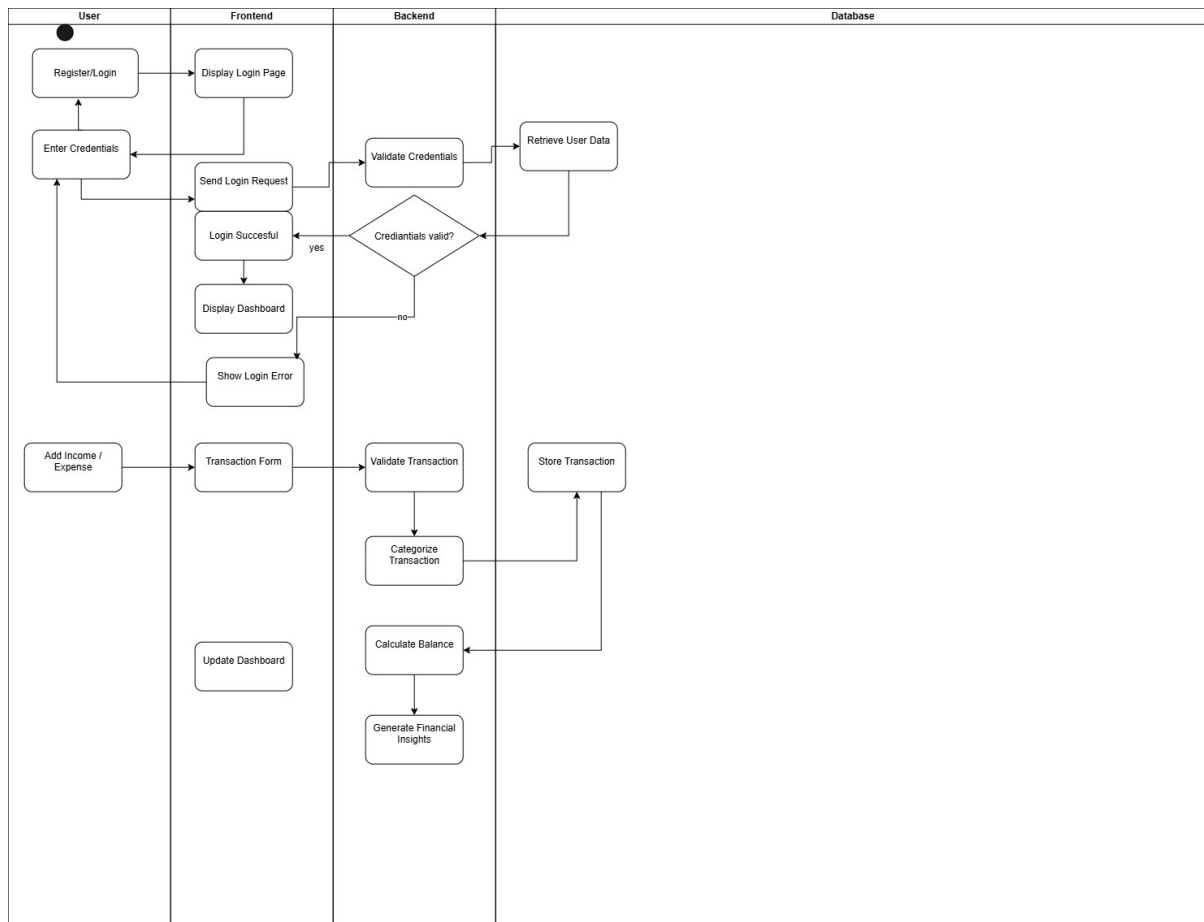


Figure 8.4: Swimlane Activity Diagram

8.5 Use Case Diagram

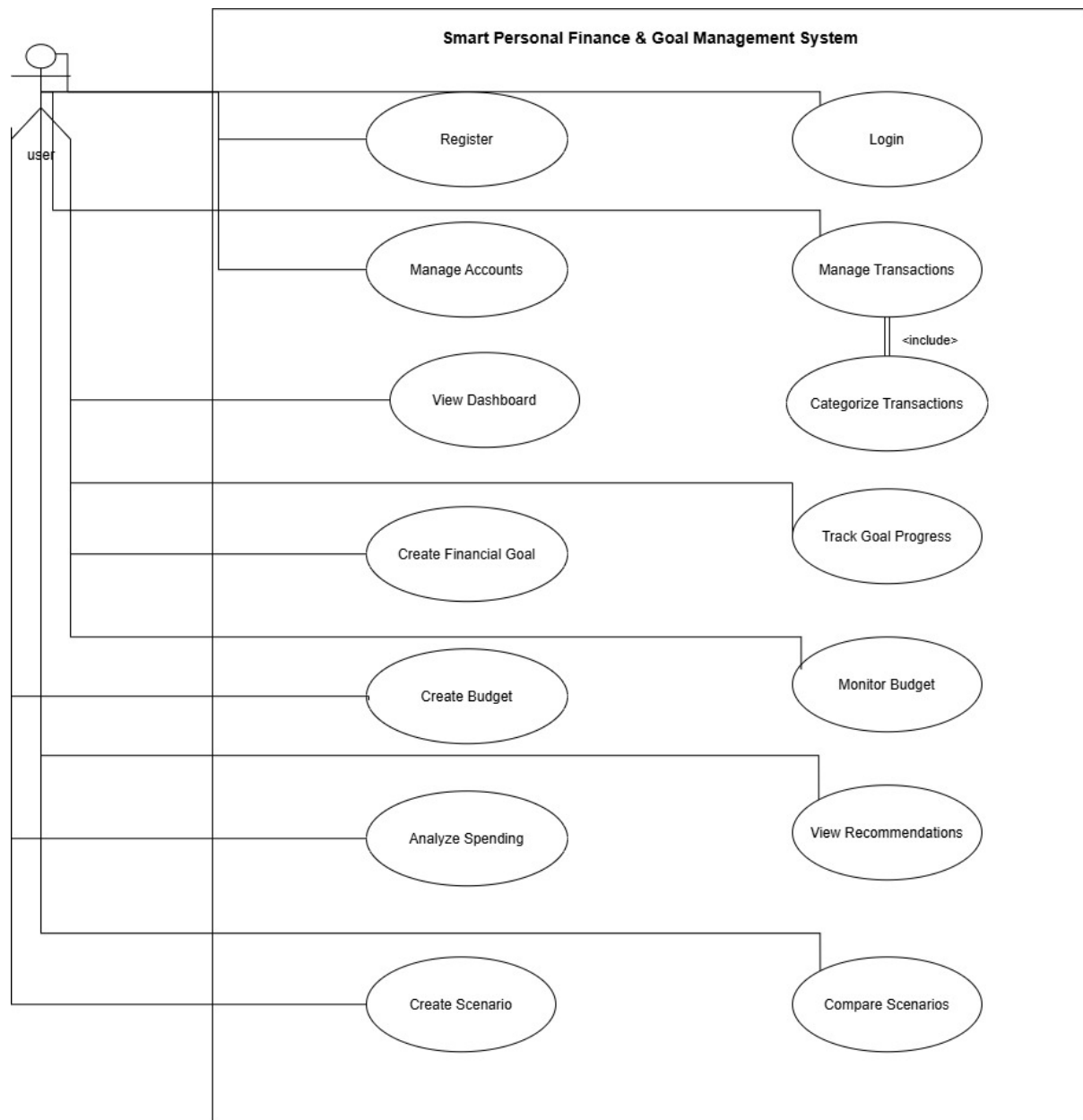


Figure 8.5: Use Case Diagram

Chapter 9

Team Responsibilities

Member	Responsibility
Riya Garg	Backend and Database Lead: REST API, Java, MySQL, JWT authentication, APIs, database design, transaction/goal/budget modules, ER diagram, API documentation and backend architecture.
Niharika	ML and Financial Intelligence Lead: Pandas/NumPy, transaction analysis, spending-habit detection, anomaly detection, forecasting, Monte Carlo simulation, scenario comparison and related algorithm documentation.
Anushka Aggarwal	Frontend/UI/UX Lead: React, Tailwind CSS, dashboard, transactions, goals, budget, charts, scenario interface, insights interface, prototype/wireframes and user flow.
Snehal Agarwal	QA/DevOps and Rules Lead: rule engine, Pytest, Hypothesis, integration/property testing, GitHub Actions, Docker, code quality, README, testing and risk documentation.
All Members	Architecture, integration, final report, presentation and final project integration.

Chapter 10

Development Plan

Period	Planned Work
Week 1	Finalize architecture, technology stack, features and responsibilities. Riya works on backend/database/ER design; Niharika prepares transaction data and financial metrics; Anushka prepares prototype and dashboard wireframe; Snehal prepares Gantt, requirements, GitHub workflow and testing plan.
Weeks 2–3	Implement MVP flow: login, dashboard, transactions, categories, budget and goals.
Weeks 4–6	Implement smart features: transaction-history analysis, behaviour analysis, anomaly detection, rule engine, Monte Carlo simulation, scenario comparison and recommendations.
Weeks 7–8	Integration, bug fixing, testing, security checks, UI/chart refinement, documentation, screenshots, final report, presentation and project demonstration.

Chapter 11

Testing Strategy

Testing is planned at multiple levels:

- **Unit testing:** individual backend functions, rules and calculations.
- **Integration testing:** interaction between frontend, backend and database.
- **Property-based testing:** validation of financial rules and calculations using Hypothesis.
- **Authentication testing:** login, protected routes and user-access control.
- **API testing:** validation of request and response behaviour.
- **Regression testing:** repeated verification after feature integration.
- **CI testing:** automated checks through GitHub Actions.

Chapter 12

Expected Outcome

The expected outcome is a working web application that combines personal-finance record keeping with financial intelligence. A user should be able to record transactions, monitor budgets and goals, inspect spending patterns, receive alerts about unusual behaviour and compare possible financial scenarios through a single dashboard.

The system is also designed to keep recommendations understandable by connecting them to financial conditions and configurable rules rather than presenting unexplained outputs.

Chapter 13

Conclusion

The Smart Personal Finance & Goal Management System integrates personal financial management, analytics and decision-support features into one application. The proposed architecture separates the frontend, backend, database and intelligence components, allowing the team to develop and test the system in parallel.

The project begins with a focused MVP and then extends it with anomaly detection, rule-based insights, forecasting, Monte Carlo simulation and scenario comparison. The combination of these modules is intended to provide a structured and explainable approach to personal financial planning.